STATISTICAL ANALYSIS OF DEFECTS OUTPUT OF PRODUCTION STERILE MEDICAL SOLUTION



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Introduction

The manufacturing of pharmaceutical drug requires controlling , monitoring & supervising of the drug, machines and operators to ensure that everything comply to the highest quality standards as **Good Manufacturing Process (GMP)** and **International Organization of standardization (ISO)** from receiving materials from suppliers until delivering the final product to the warehouse.

Research Important

Defects during drug manufacturing one of the most common issues that faces the pharmaceutical companies because of the following;

1-HIGH COST OF EVERY UNIT OF DRUG because of high standards required for manufacturing of the drug so every unit cost about 5-150 pound depend on the classification of the area and type of the drug.

2- **TIME CONSUMING** because of the drug defects are out of production plan.

3-**POOR QUALITY OF THE MANUFACTURER** reflects the poor customer satisfaction that affects sales.

3- **LEGAL ACCOUNTABILITY & PRODUCT RECALL,** if the major defects appear in the market by mistake.

Research variables

Study the relation between of the **total goods produced (y – dependent variable)** with defects output (x-independent variable) in factory for the production of sterile medical solution from January 2017 until June 2019.

The defects output of manufacturing of sterile medical solution classified into three main categories:

1- **Manufacturing defects (X1) dependant variable**– which include defects that arise from design of the bottle pack from plastic granules, fill it with the drug and seal it with cap, known as **Blow fill seal (BFS).**

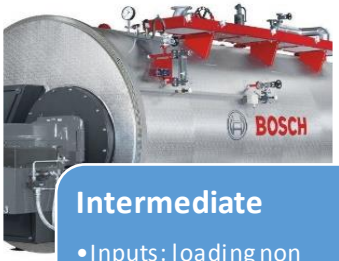
2-**Autoclave defects (X2) dependant variable** – which include defects that arise from the handling of BP by workers from loading BP into trays to be sterilized by autoclave then unloading BP into belt conveyor.

3-Packaging *defects (X3) dependant variable* – which include defects that arise from putting the labels into the sterilized BP, wrapping it with film & putting into specialized carton.



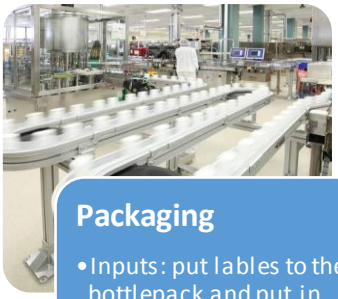
Manufacturing

- Inputs: Raw material , serivces & machines.
- outputs: non sterile bottle pack



Intermediate

- Inputs: loading non sterile bottlepack into trays.
- outputs : unloading sterile bottlepacj into trays.



Packaging

- Inputs: put lables to the bottlepack and put in cartons.
- outputs : final product deilived to warehouse.

Data analysis

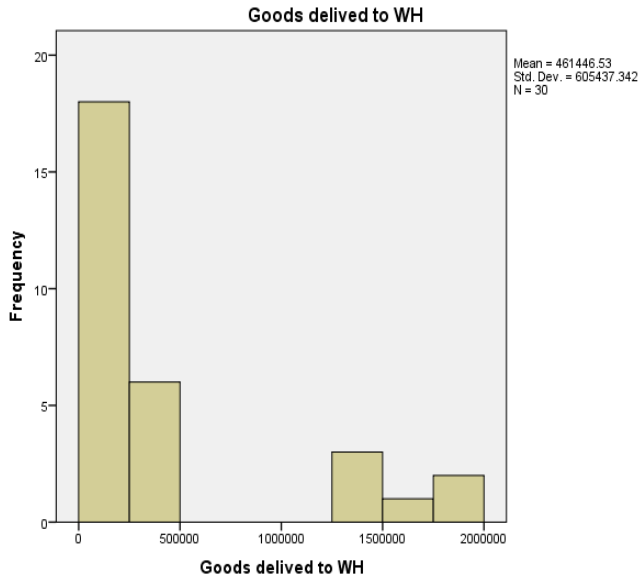
First : Discriptive stattistics

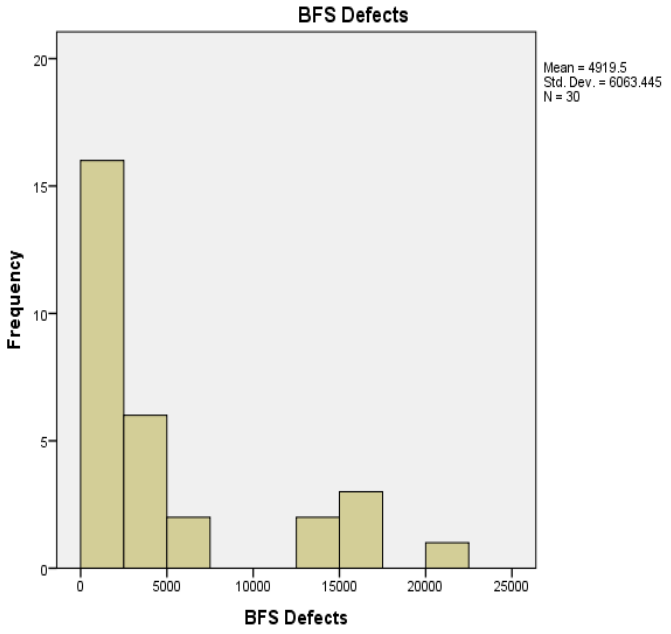
Statistics

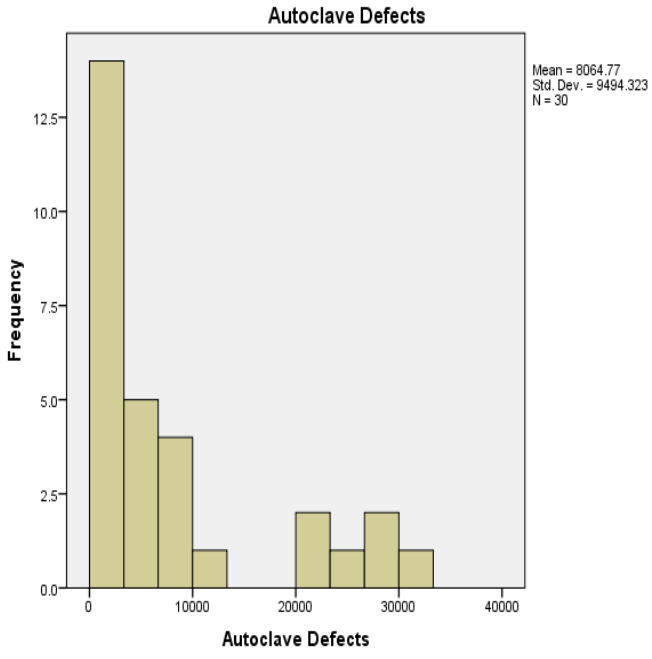
		Goods delived to WH	BFS Defects	Autoclave Defects	Packaging defects
N	Valid	30	30	30	30
	Missing	0	0	0	0
Mean		461446.53	4919.50	8064.77	428.33
Median		157109.50	2228.00	4547.50	132.00
Mode		21836[a]	179[a]	239[a]	103
Std. Deviation		605437.342	6063.445	9494.323	610.615
Variance		3.666E+11	36765362.88	90142162.94	372850.920
Range		1853696	20148	31077	1917
Minimum		21836	179	239	8
Maximum		1875532	20327	31316	1925

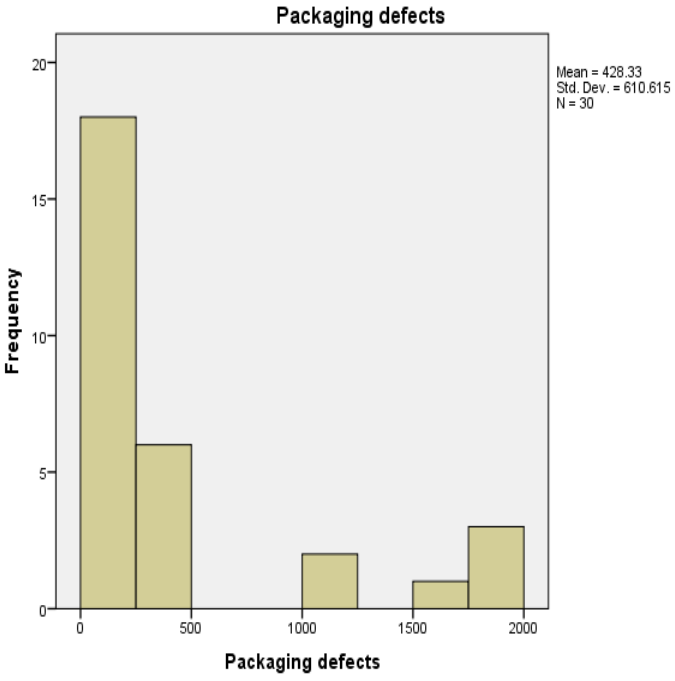
a. Multiple modes exist. The smallest value is shown

Histogram









Second : Correlation statistics:

Correlations

			Goods delived to WH	BFS Defects	Autoclave Defects	Packaging defects
Spearman's rho	Goods delived to WH	Correlation Coefficient	1.000	.939**	.895**	.924**
		Sig. (2-tailed)	.	.000	.000	.000
		N	30	30	30	30
	BFS Defects	Correlation Coefficient	.939**	1.000	.951**	.971**
		Sig. (2-tailed)	.000	.	.000	.000
		N	30	30	30	30
	Autoclave Defects	Correlation Coefficient	.895**	.951**	1.000	.948**
		Sig. (2-tailed)	.000	.000	.	.000
		N	30	30	30	30
	Packaging defects	Correlation Coefficient	.924**	.971**	.948**	1.000
		Sig. (2-tailed)	.000	.000	.000	.
		N	30	30	30	30

**. Correlation is significant at the 0.01 level (2-tailed).

Correlation interpretetion:

1- The Correlation Coefficient between **Goods delivered to WH (Y factor)** & **BFS Defects (X_1) factor** equals (**0.939**) this means the correlation is **very strong with direct relation** and the correlation is **significant** because margin of error is less than 0.05.

2- The Correlation Coefficient between **Goods delivered to WH (Y) factor** & **Autoclave Defects (X_2) factor** equals (**0.895**) this means the correlation is **strong with direct relation** and the correlation is **significant** because margin of error is less than 0.05.

3- The Correlation Coefficient between **Goods delivered to WH (Y) factor** & **Packaging Defects (X_3) factor** equals (**0.924**) this means the correlation is **very strong with direct relation** and the correlation is **significant** because margin of error is less than 0.05.

Third: Regression statistics

I-simple form

1- Goods deived to WH (Y) vs BFS defects (X_1)

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	BFS Defects[b]	.	Enter

a. Dependent Variable: Goods delived to WH

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.985[a]	.971	.970	104829.979

a. Predictors: (Constant), BFS Defects

Data interpretation

1-The correlation equals (0.985), means the correlation between **Goods deived to WH (Y) & BFS defects (X_1)** is **very strong with direct relation**.

2-Regression square equals (0.971), means the weight of BFS defects with error represent about **97.1 %** of Goods delivered to WH varation.

3-Adjusted regression square equals (0.970), means the weight of BFS defects without error represent about **97 %** of Goods delivered to WH varation.

So, Error equals (0.1) %.

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1.032E+13	1	1.032E+13	939.309	.000[b]
	Residual	3.077E+11	28	10989324552		
	Total	1.063E+13	29			

a. Dependent Variable: Goods delived to WH

b. Predictors: (Constant), BFS Defects

- The significance is less than (0.05) ,i.e. **significant**, so **this model is suitable for regression model**

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-22605.858	24814.442		-.911	.370
	BFS Defects	98.395	3.210	.985	30.648	.000

a. Dependent Variable: Goods delived to WH

- The cosntant (a) equals (0.37) ,i.e. **insignificane** because > 0.05, & logic because adjusted R sqaure > 75 %.
- BFS defects (b_1) equals (0.00) ,i.e. **significance**.

So, the simple regression equation is

$$Y = a + b_1X_1 + E$$

$$Y = 98.395\,X_1 + 0.1\%\ error\ range$$

1- Goods deived to WH (Y) vs Autoclave defects (X_2)

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Autoclave Defects[b]	.	Enter

a. Dependent Variable: Goods delived to WH

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.980[a]	.961	.960	121401.113

a. Predictors: (Constant), Autoclave Defects

Data interpretation

1-The correlation equals (0.98), means the correlation between **Goods deived to WH (Y) & Autoclave defects (X_2)** is **very strong with direct relation**.

2-Regression square equals (0.961), means the weight of autoclave defects with error represent about **96.1 %** of Goods delivered to WH varation.

3-Adjusted regression square equals (0.970), means the weight of autoclave defects without error represent about **96 %** of Goods delivered to WH varation.

So, Error equals (0.1) %.

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1.022E+13	1	1.022E+13	693.259	.000[b]
	Residual	4.127E+11	28	14738230205		
	Total	1.063E+13	29			

a. Dependent Variable: Goods delived to WH

b. Predictors: (Constant), Autoclave Defects

- The significance is less than (0.05) ,i.e. significant, so this **model is suitable for regression model**

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-42749.187	29291.085		-1.459	.156
	Autoclave Defects	62.518	2.374	.980	26.330	.000

a. Dependent Variable: Goods delived to WH

- The cosntant (a) equals (0.156) ,i.e. **insignificane** because > 0.05, & logic because adjusted R sqaure > 75 %.
- Autoclave defects (b_2) equals (0.00) ,i.e. **significance**.

So, the simple regression equation is

$$Y = a + b_2X_2 + E$$

$$Y = 62.518\ X2 + 0.1\%\ error\ range$$

1- Goods deived to WH (Y) vs Packaging defects (X_3)

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Packaging defects[b]	.	Enter

a. Dependent Variable: Goods delived to WH

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.983[a]	.966	.964	114119.737

a. Predictors: (Constant), Packaging defects

Data interpretation

1-The correlation equals (0.983), means the correlation between **Goods deived to WH (Y) & Packaging defects (X_3)** is **very strong with direct relation.**

2-Regression square equals (0.966), means the weight of packaging defects with error represent about **96.6 %** of Goods delivered to WH varation.

3-Adjusted regression square equals (0.964), means the weight of autoclave defects without error represent about **96.4 %** of Goods delivered to WH varation.

So, Error equals (0.2) %.

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1.027E+13	1	1.027E+13	788.234	.000[b]
	Residual	3.647E+11	28	13023314432		
	Total	1.063E+13	29			

a. Dependent Variable: Goods delived to WH

b. Predictors: (Constant), Packaging defects

The significance is less than (0.05) ,i.e. **significant**, so ***this model is suitable for regression model***

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	44093.367	25594.726		1.723	.096
	Packaging defects	974.365	34.705	.983	28.076	.000

a. Dependent Variable: Goods delived to WH

The cosntant (a) equals (0.096) ,i.e. **insignificane** because >0.05, & logic because adjusted R sqaure > 75 %.
Packaging defects (b_3) equals (0.00) ,i.e. **significance**.

So, the simple regression equation is

$$Y = a + b_3X_3 + E$$

$$Y = 974.365\ X3 + 0.2\%\ error\ range$$

II-Multiple regression Form

Goods deived to WH (Y) vs BFS defects (X_1), Autoclave defects (X_2) & Packaging defects (X_3)

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Packaging defects, BFS Defects, Autoclave Defects[b]	.	Enter

a. Dependent Variable: Goods delived to WH

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.992[a]	.984	.982	81832.894

a. Predictors: (Constant), Packaging defects, BFS Defects, Autoclave Defects

Data interpretation

1-The correlation equals (0.992), means the correlation between **Goods deived to WH (Y) & BFS defects (X1), Autoclave defects (X2) & Packaging defects (X3)** is very strong with direct relation.

2-Regression square equals (0.984), means the weight of BFS defects (X1), Autoclave defects (X2) & Packaging defects (X3) with error represent about ***98.4 %*** of Goods delivered to WH varation.

3-Adjusted regression square equals (0.982), means the weight of BFS defects (X1), Autoclave defects (X2) & Packaging defects (X3) without error represent about ***98.2 %*** of Goods delivered to WH varation.

So, Error equals (0.2) %.

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1.046E+13	3	3.485E+12	520.460	.000[b]
	Residual	1.741E+11	26	6696622477		
	Total	1.063E+13	29			

a. Dependent Variable: Goods delived to WH

b. Predictors: (Constant), Packaging defects, BFS Defects, Autoclave Defects

The significance is less than (0.05) ,i.e. **significant**, so **this model is suitable for regression model**

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-15595.569	21951.748		-.710	.484
	BFS Defects	44.019	12.430	.441	3.541	.002
	Autoclave Defects	16.292	8.021	.255	2.031	.053
	Packaging defects	301.410	130.542	.304	2.309	.029

a. Dependent Variable: Goods delived to WH

- The cosntant (a) equals (0.484) ,i.e. **insignificane** because > 0.05, & logic because adjusted R sqaure > 75 %.
- BFS defects (b_1) equals (0.02) ,i.e. **significance**
- Autoclave defects (b_2) equals (0.53) ,i.e. **insignificance**.
- Packaging defects (b_3) equals (0.29) ,i.e. **significance**

So the multiple regression equation is

$$Y = \cancel{a} + b_1X_1 + \cancel{b_2X_2} + b_3X_3 + E$$

$$Y = 44.019\,X1 + 301.410\,X3 + 0.2\%\ error\ range$$

Interpretion of insignificance of Autoclave defects (b_2) form Correlation table

Correlations

			Goods delived to WH	BFS Defects	Autoclave Defects	Packaging defects
Spearman's rho	Goods delived to WH	Correlation Coefficient	1.000	.939**	.895**	.924**
		Sig. (2-tailed)	.	.000	.000	.000
		N	30	30	30	30
	BFS Defects	Correlation Coefficient	.939**	1.000	.951**	.971**
		Sig. (2-tailed)	.000	.	.000	.000
		N	30	30	30	30
	Autoclave Defects	Correlation Coefficient	.895**	.951**	1.000	.948**
		Sig. (2-tailed)	.000	.000	.	.000
		N	30	30	30	30
	Packaging defects	Correlation Coefficient	.924**	.971**	.948**	1.000
		Sig. (2-tailed)	.000	.000	.000	.
		N	30	30	30	30

**. Correlation is significant at the 0.01 level (2-tailed).

1-The Correlation coefficient between Autoclave defects (X2) & BFS defects (X1) equals (0.951), means very strong with direct relation and the correlation is **significant** because margin of error is less than 0.05.

2-The Correlation between Autoclave defects (X2) & Goods Deliverd to WH (Y) equals (**0.895**) this means the correlation is **strong with direct relation** and the correlation is **significant** because margin of error is less than 0.05, while the other factors BFS defects (X1) & Packaging defects (X3) have **strong with direct relation with** Goods Deliverd to WH (Y).

So, Autoclave defects (X2) would't be used as factor in Multiple regression form

Multiple regression form
Goods deived to WH (Y) vs BFS defects (X1) & Packaging defects (X3)

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Packaging defects, BFS Defects[b]	.	Enter

a. Dependent Variable: Goods delived to WH

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.990[a]	.981	.980	86439.689

a. Predictors: (Constant), Packaging defects, BFS Defects

Data interpretation

1-The correlation equals (0.990), means the correlation between **Goods deived to WH (Y) & BFS defects (X_1), & Packaging defects (X_3)** is ***very strong with direct relation.***

2-Regression square equals (0.981), means the weight of BFS defects (X_1) & Packaging defects (X_3) with error represent about ***98.1 %*** of Goods delivered to WH varation.

3-Adjusted regression square equals (0.980), means the weight of BFS defects (X_1) & Packaging defects (X_3) without error represent about ***98 %*** of Goods delivered to WH varation.

So, Error equals (0.1) %.

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1.043E+13	2	5.214E+12	697.845	.000[b]
	Residual	2.017E+11	27	7471819836		
	Total	1.063E+13	29			

a. Dependent Variable: Goods delived to WH

b. Predictors: (Constant), Packaging defects, BFS Defects

- The significance is less than (0.05) ,i.e. significant, so this **model is suitable for regression model.**

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1419.230	21432.778		.066	.948
	BFS Defects	55.095	11.799	.552	4.669	.000
	Packaging defects	441.221	117.164	.445	3.766	.001

a. Dependent Variable: Goods delived to WH

- The cosntant (a) equals (0.948) ,i.e. **insignificane** because > 0.05, & logic because adjusted R sqaure > 75 %.
- BFS defects (b_1) equals (0.00) ,i.e. **significance**
- Packaging defects (b_3) equals (0.001) ,i.e. **significance**

So the multiple regression equation is

$$Y = a + b_1X_1 + b_3X_3 + E$$

$$Y = 55.095\,X1 + 441.221\,X3 + 0.1\%\ error\ range$$

Conclusion

- By studying the correlation between Goods delived to WH (Y) with BFS defects (X_1), Autoclave defects (X_2)& Packaging defects (X_3) it is found the relation is intersted & significant.

The following table represent the priority of the Correaltion between Goods delived to WH (Y) & (X) factors :

S.N	Items / X factors	Level of Relation	Relation direction	Margin of error
1	BFS defects (X_1)	Very strong	Direct	Significance
2	Packaging defects (X_3)	Very strong	Direct	Significance
3	Autoclave defects (X_2)	Strong	Direct	Significance

- And By studying the regression simple & multiple form between Goods delived to WH (Y) vs BFS defects (X_1), Autoclave defects (X_2) & Packaging defects (X_3) are suitable for regression model.

The following table represent the coefficent signifacne between Each (X) factor in simple & multiple regression with (Y) factor:

S.N	Items / X factors	Simple Regression	Multiple Regression
1	BFS defects (X_1)	Significant	Significant
2	Autoclave defects (X_2)	Significant	Insignifacnt
3	Packaging defects (X_3)	Significant	Significant

- It has been clarified in Correlation coeffiecnt between Autoclave defects & BFS defects are very strong with direct relation.
- **This is logic because the Stage Autoclave after stage of BSF**

The best form of multiple regression is between
Goods delivered to WH (Y) vs BFS defects (X1) & Packaging defects (X3) `
which has the most impact on producitiviy of sterile medical solution & can be presents in the following eqution:

$$Y = 55.095\,X1 + 441.221\,X3 + 0.1\%\ error\ range$$

Recommendation

In order to save **time ,money & effort** in order to decrease defects of the manufacturing of sterile medical solution, the study recommend the following **respectively**:

1-There is **no significant to deal with the defects of Autoclave** because it's affected by the defects of BFS stage.

2-Also it recommend to make root cause of **analysis of BFS stage to determine the cause of its defects** (which can be change the required spare part of the machine of BFS) & by decreasing the defects pf BFS, the defects of autoclave will be decreased respectively.

3-Also it recommend **determine the causes of the Packaging defects** to determine which machine &or machines make the highest rate of defects.

All these Recommendations would decrease the defects of production sterile medical solution and subseculty enhance the productivity.

Thank you